

Problem Set 2
Intermediate Microeconomics
Spring 2023

1. Chapter 8's Exercises [3.17](#), [4.2](#), [4.4](#)

Note: The cost function in Exercise 4.4 should be $C = q - q^2 + q^3$

2. Leonard's firm produces granola bars. His short run cost function is

$$C = \frac{1}{3}q^3 + 20$$

- a. Assuming the market for granola bars is competitive, derive Leonard's short-run firm supply function.
- b. Suppose there are 100 identical firms (including Leonard's) producing granola bars. What is the short-run market supply?
- c. Suppose the market demand function is $Q^D = \frac{1600}{p^{0.5}}$. What is the equilibrium price in the short-run? How many granola bars does each firm produce in the short-run equilibrium?

a. $C = \frac{1}{3}q^3 + 20$

$$MC = q^2 \quad AVC = \frac{1}{3}q^2$$

$$P = MC = q^2$$

$$\Rightarrow P = MC > AVC \rightarrow \text{SR supply fn.} : q = \sqrt{P}, P \geq 20$$

b. SR market supply $\Rightarrow Q^S = 100\sqrt{P}$

c. $Q^D = Q^S \Rightarrow \frac{1600}{p^{0.5}} = 100 p^{0.5}$

$$1600 = 100 p$$
$$p = 16$$

$$q = 4$$

$$3.17 \quad C = q^3 - 36q^2 + 490q + 1,000$$

$$\pi = R - C$$

$$\Rightarrow 250q - (q^3 - 36q^2 + 490q + 1,000)$$

$$MC = 3q^2 - 72q + 490$$

$$P = MC \Rightarrow 250 = 3q^2 - 72q + 490$$

$$0 = 3q^2 - 72q + 240$$

$$0 = q^2 - 24q + 80$$

$$0 = (q - 4)(q - 20)$$

$$q = 4 \vee 20$$

$$MC'$$

$$= 6q - 72$$

$$MC' > 0$$

(20)

$$\pi = 250 \times 20 - (20^3 - 36 \times 20^2 + 490 \times 20 + 1,000)$$

$$AVC = q^2 - 36q + 490$$

$$\downarrow AVC = 2q - 36$$

$$q = 18$$

operate

4.2

$$4.4 \quad C = q - q^2 + q^3 + q.$$

$$Q = 24 - p$$

$$MC = 1 - 2q + 3q^2$$

$$AC = 1 - q + q^2$$

$$1 - 2q + 3q^2 = 1 - q + q^2$$

$$2q^2 - q = 0$$

$$q(2q - 1) = 0$$

$$q = 0 \vee \frac{1}{2}$$

$$p = 1 - 2 \times \frac{1}{2} + \frac{3}{4}$$

$$= \frac{3}{4}$$

$$p^* = \frac{3}{4}$$

$$q^* = \frac{1}{2}$$

$$Q = 24 - 0.75$$

$$= 23.25$$

$$n^* = \frac{23.25}{0.5} = 46.5$$

$$C = 2q - q^2 + q^3$$

$$MC = 2 - 2q + 3q^2$$

$$AC = 2 - q + q^2$$

$$q^* \text{ const.}$$

$$p^* = 2 - 2 \times \frac{1}{2} + 0.75 = 1.75$$

$$n^* = 44.5$$

$$201 \times 25000 \\ 125000$$

$$4.19 \quad C = 40q - q^2 + 0.01q^3$$

$$Q = 5,000 - 200p$$

$$AC = 40 - q + 0.01q^2$$

$$MC = 40 - 2q + 0.03q^2$$

$$40 - q + 0.01q^2 = 40 - 2q + 0.03q^2$$

$$0 = 0.02q^2 - q$$

$$5,000 - 200 \times 15$$

$$5,000 - 3,000$$

$$= 2,000$$

$$0 = q(0.02q - 1)$$

$$q = 0 \vee 50$$

$$p = MC$$

$$q^* = 50$$

$$p^* = 15$$

$$n^* = 40$$

$$\frac{40}{2,000} \\ \frac{2,000}{40} \\ =$$

$$\Rightarrow 40 - 2 \times 50 + 0.03 \times 500$$

$$40 - 100 + 75$$

$$-60 + 75 = 15$$

$$Y = ZK$$

$$a. \Delta K = sY - \delta K$$

$$\Rightarrow \Delta K = sZK - \delta K$$

$$\Rightarrow \frac{\Delta K}{K} = sZ - \delta = \frac{\Delta Y}{Y}$$

if $sZ > \delta \Rightarrow$ income per person
grows indefinitely

$$3.1) C(q) = a + bq + cq^2 + dq^3$$

$$MC(q) = b + 2cq + 3dq^2$$

$$AC(q) = \frac{a}{q} + b + cq + dq^2$$

$$\Rightarrow P = MC = b + 2cq + 3dq^2$$